ARYAN MEENA

Data Scientist | Python, SQL, PyTorch, Machine Learning
araj7042@gmail.com | (857) 425-6779 | LinkedIn | GitHub | Portfolio | Boston, MA (open to relocation)

EDUCATION

Master of Science in Applied Data Analytics - Boston University | GPA 3.75/4.0 **May 2026**
Bachelor of Technology in Electronics and Communication Engineering - Indian Institute of Space Science and Technology **May 2023**
Relevant Coursework: Advanced Machine Learning, Foundations of Machine Learning, Big Data Analytics, Data Mining
Honors: First runner-up of 40+ teams, BU MedAI Hackathon (team lead); 3D PET model at validation MAE 7.31 Centiloid (r 0.968)

PROFESSIONAL EXPERIENCE

Research Assistant, Data Science and Quantitative Finance, Boston University | *Boston, MA* **Sep 2025 – May 2026**
Skills – Python (pandas, NumPy), SQL, scikit-learn, time series, statistical testing, Git

- Traced a 15-year Dollar Index trading signal to bad opening prices in a free data feed by checking the feed's data quality against exchange data in Python. **Published as first author in Risks (MDPI).**
- Forecast 5-day currency volatility with time series and machine learning models (regression, random forest) at out-of-sample R^2 0.25 vs a 0.05 noise ceiling.
- Caught 5 feature engineering bugs before model fitting, one of which would have leaked 615 false zeros into the data.
- Rewrote the NumPy backtest engine behind 1,312 strategy configurations to run 17x faster with identical results and kept every table reproducible under Git version control.
- Wrote the SQL and Python behind a live research dashboard on a 12-table relational database (484K monthly rows) so non-specialists can check every result without code.

Graduate Teaching Assistant, Boston University | *Boston, MA* **Jan 2026 – May 2026**
Skills – Python, scikit-learn, PyTorch, statistics

- Held weekly office hours for 60+ graduate students training as data scientists in Advanced Machine Learning and Data Science with Python, debugging student code and explaining model results in plain language.

Machine Learning Research Intern, ISRO Space Applications Centre | *Ahmedabad, India* **Jun 2022 – Jul 2022**
Skills – PyTorch, U-Net, DeepLabV3+, QGIS

- Trained U-Net and DeepLabV3+ deep learning models in PyTorch to segment roads in 6,723 aerial images, and traced a drop in accuracy on Indian satellite imagery to a sensor resolution and metadata mismatch.

PROJECTS

Hospital Readmission Risk Model | *Python, LightGBM, PyTorch, Tableau* **Live dashboard**

- **Beat a published 30-day readmission benchmark (AUC 0.689 vs 0.664)** with a gradient-boosted trees clinical prediction model on 101,766 electronic health record (EHR) encounters.
- Ran an algorithmic fairness audit by age, race and sex (no race or sex gap, stable calibration), recommended age-specific thresholds after AUC fell from 0.733 to 0.631 across age bands, and wrote TRIPOD+AI model documentation.
- Published a care-team dashboard in Tableau where the top 5% of flagged patients had 3x the readmissions of a random 5%.

JudgeGuard: LLM Evaluation | *Python, FastAPI, Docker, Hugging Face* **Live app**

- Tested a small open-source large language model (Qwen3-4B) as an LLM-as-a-judge for generative AI answers, choosing the answer backed by the source in **96 of 100 test pairs** (95% CI 92–99%).
- Deployed the judge as a REST API (FastAPI) in Docker and a live Hugging Face app, with 150 automated tests in GitHub Actions CI.

WikiFlow: Editor Dropout Prediction | *PySpark, Google Cloud Dataproc, BigQuery* **Springer paper**

- Processed **60M Wikipedia edit events** (~25 GB) through a Spark data pipeline on a 5-node Google Cloud cluster to predict which of 1.16M new editors drop out, in about 2 hours for roughly $2.50.
- Reached AUC 0.909 (precision and recall 0.87) with a distributed neural network, 6.4 points above logistic regression.

PUBLICATIONS

Meena and Pinsky. Two Blind Spots in Backtest Validation. *Risks (MDPI), 14(11), 247, 2026.*
Meena et al. Accelerated CNN Training with Genetic Algorithm Optimization. *IEEE IATMSI, 2024.*

TECHNICAL SKILLS

Programming and ML: Python (pandas, NumPy, scikit-learn), SQL, R, PyTorch, LightGBM, XGBoost, deep learning, NLP, LLM evaluation
Statistics: regression, hypothesis testing, bootstrap, cross-validation, calibration, time series forecasting
Data, Cloud and Tools: Spark (PySpark), PostgreSQL, AWS (S3, Glue, Athena), GCP (BigQuery), Tableau, Power BI, Git, Docker, CI/CD